

Kumar M

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PROFESSIONAL SUMMARY

Senior Embedded Systems Engineer with 7+ years of experience in firmware development, silicon bring-up and device driver engineering across aerospace, automotive and semiconductor platforms. Delivered production firmware for ISRO satellite missions and Microchip's PICkit 5/ICD 5 debugger tools. Proficient in bare-metal C, FreeRTOS, device driver development and Python-based silicon validation automation.

TECHNICAL SKILLS

- **Languages:** Embedded C, Python, C++, Data Structures & Algorithms.
- **Hardware:** ARM Cortex-M (M0+/M4/M7), ARM Cortex-A53 (ZU7EV PS), FPGA (Xilinx Zynq Ultra-Scale+ ZU7EV, ZC706), SoC Bring-up, JTAG/SWD, Oscilloscopes, Logic Analysers.
- **Interfaces:** SPI, I2C, UART, PCIe, GPIO, DMA, ADC.
- **OS / RTOS:** FreeRTOS (task, queue, semaphore, mutex, ISR-safe APIs), Zephyr RTOS (Device Tree), Linux (kernel modules, platform drivers, POSIX – pthreads, IPC), Bare Metal.
- **Driver Model:** Linux platform driver, ioremap/ioread32/iowrite32, devm_* APIs, sysfs interface, module_platform_driver.
- **Security:** Root of Trust, secure boot chain of trust, OTP and TF-M concepts.
- **Tools:** pyOCD, Keil, IAR, GCC, Jenkins CI/CD, Git, Vivado, Vitis, MPLAB X.

EXPERIENCE

Microchip Technology

May 2024 – Present

Senior Engineer - I, Development Tools

Bengaluru

- Senior firmware developer for **PICkit 5 and ICD 5 Debugger Tools firmware** – implemented hardware-layer abstraction (HAL) in Embedded C on a FreeRTOS architecture with priority-based preemptive scheduling and inter-task queue communication.
- Developed **CMSIS-Pack firmware** and **Flash Loaders** for ARM Cortex-M0+ families, enabling Tier-1 ecosystem partners (Keil, IAR) with validated silicon bring-up artifacts.
- Architected a scalable **Silicon Enablement Framework** using Python (pyOCD/CMSIS-DAP v2) to automate deployment and register-level verification of Flash Programming Algorithms across **800+ MCU variants**.
- Reduced validation-to-release cycle time by **60%** through a generic high-performance automated testing pipeline integrated with Jenkins CI/CD.
- Designed structured test plans covering power-on sequencing, register-level verification and JTAG/SWD debug validation – directly mirroring first-silicon bring-up methodology.

Bosch Global Software Technologies

September 2022 – May 2024

Senior Engineer - I, Customer Engineering - BMW

Bengaluru

- Engineered **safety-critical ECU firmware** for BMW automotive platforms – PI control loops, real-time thermal and power management, ISO 26262 compliance.
- Implemented and verified functional safety modules at system level, ensuring AUTOSAR-compliant firmware delivery.
- Reduced RAM footprint and HEX binary size via **custom linker script optimisation** – section placement, dead code elimination, load/run address management. (1st Place, BU Hackathon).

- **Satellite OBC Bring-up:** Led full-cycle board bring-up and schematic verification for ISRO INS-2TD and Pixxel ANAND On-Board Computers – power sequencing, JTAG boundary scan, clock verification, and peripheral driver validation on first silicon.
- **FPGA Bring-up:** Board bring-up and firmware development for Xilinx Zynq UltraScale+ ZU7EV and ZC706 platforms – FSBL, peripheral integration, Vivado/Vitis toolchain.
- **Satellite TTC Systems:** Designed and implemented ground station and on-board Telemetry & Telecommand firmware with high-reliability UART/I2C drivers under strict real-time constraints.
- Engineered bare-metal reaction wheel control drivers for Pixxel LEO satellites.
- **ISRO Recognition Award:** Accolades from LEO Directors for INS-2TD mission contributions.

TECHNICAL PROJECTS

ARM Cortex-M RTOS Scheduler

github.com/kumar-mohan-s/nano-sched

- Priority-based preemptive scheduler for ARM Cortex-M4 – TCB management, PSP context switching via PendSV, round-robin for equal-priority tasks.
- Implements full context save/restore: hardware exception frame (R0 – R3, R12, LR, PC, xPSR) plus manual STMDB/LDMIA for callee-saved R4 – R11.

Silicon Enablement Tool (pyOCD)

github.com/kumar-mohan-s/pydap-link

- Python utility using pyOCD/CMSIS-DAP v2 for automated core detection, ROM table parsing, and Flash programming sequence validation across ARM SoC platforms.
- Foundation of the 800-variant Silicon Enablement Framework deployed at Microchip Technology.

Edge ML Inference on ARM Cortex-M7

github.com/kumar-mohan-s/edge-ml-nsnet2-cortex-m7

- Exported NSNET2 speech enhancement model to ONNX and verified inference correctness on host (macOS) with real audio samples.
- Porting to ARM Cortex-M7 (PIC32CZ-CA90) with quantised weight mapping to TCM/SRAM and real-time I2S audio interface – demonstrates embedded ML deployment on resource-constrained silicon.

SAMC21 JTAG TAP Controller

github.com/kumar-mohan-s/samc21-jtag-tap

- Implemented IEEE 1149.1 JTAG TAP state machine on ARM Cortex-M0+ (SAMC21) using GPIO bit-bang – reads JTAG IDCODE from target silicon (PIC32CZ-CA90).

TALKS & PRESENTATIONS

AI Hardware Wars: What Embedded Engineers Need to Know

2026

[Microchip Software Summit], Bengaluru

- Presented competitive landscape of AI compute silicon – NVIDIA CUDA ecosystem, Google TPU, Graphcore IPU, and Cerebras WSE – drawing from the landmark “AI+HW 2035” paper (LeCun, Young et al.).
- Examined the shift from raw compute to intelligence-per-joule and implications for embedded and firmware engineers at the edge.

EDUCATION

Bachelor of Engineering (Electronics & Communication)

2014 – 2018

Anna University, Chennai